Agents for Impact

Hackathon



RescueBroker

Surplus food to the right pantry, under a clock

Table 17

Challenge 2: Resilient Food Equity & Surplus Broker · Track 1, retail and grocer rescue, Chicago

A matching problem under a clock

Ten crates of spinach. Six hours left. Who can take it?



What just became available

10 crates of spinach



Who could use it

Recipients who need it



What can they store

Cold, room, freezer



How long does it have

6 hours on the clock

Four questions, in the same minute. By the time a human answers them, the food is gone.

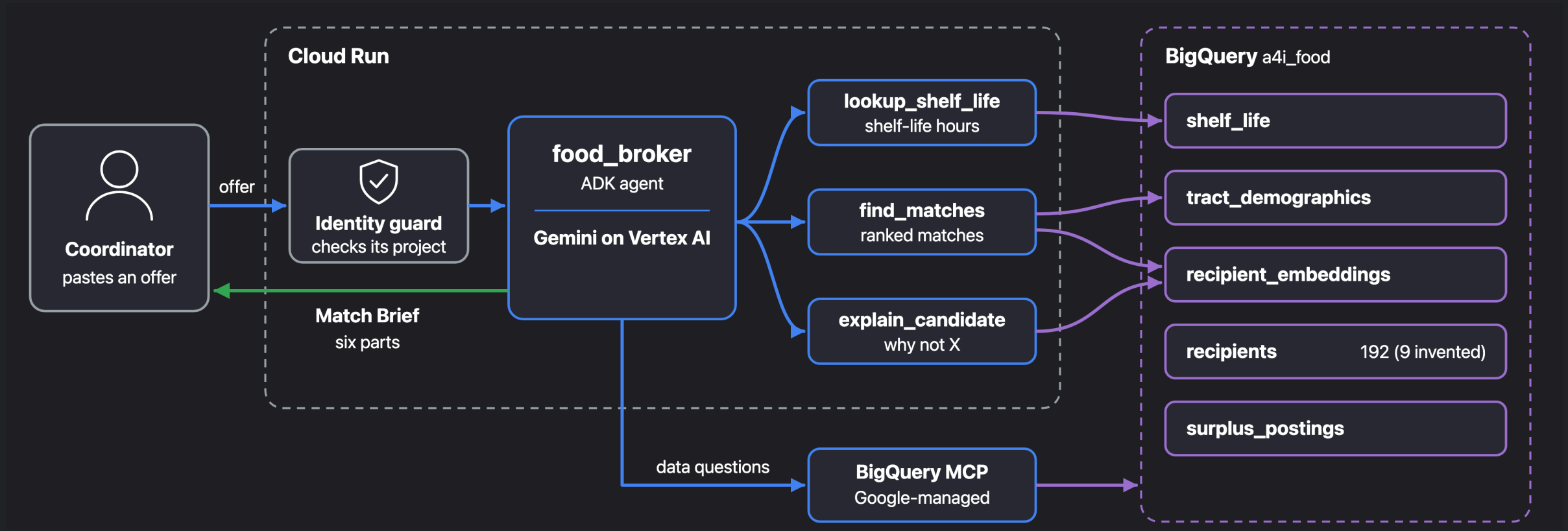
From a text message to a call-ready brief



Built on

Gemini on Vertex AI | ADK agent | BigQuery vector search | BigQuery MCP server | Cloud Run

How the pieces fit



Meaning alone picks the wrong pantry

Spinach offer. The food has 6 hours.

The 3 closest by meaning

- ✗ Needs 19 hours to collect
- ✗ Needs 12 hours to collect†
- ✗ The decoy†: no fridge, 12 hours

What we pick instead

- ✓ Keeps it cold
- ✓ Collects in time
- ✓ 2 to 3 hours to spare

Why we search 80, not 15

Top 15 → **1 viable**

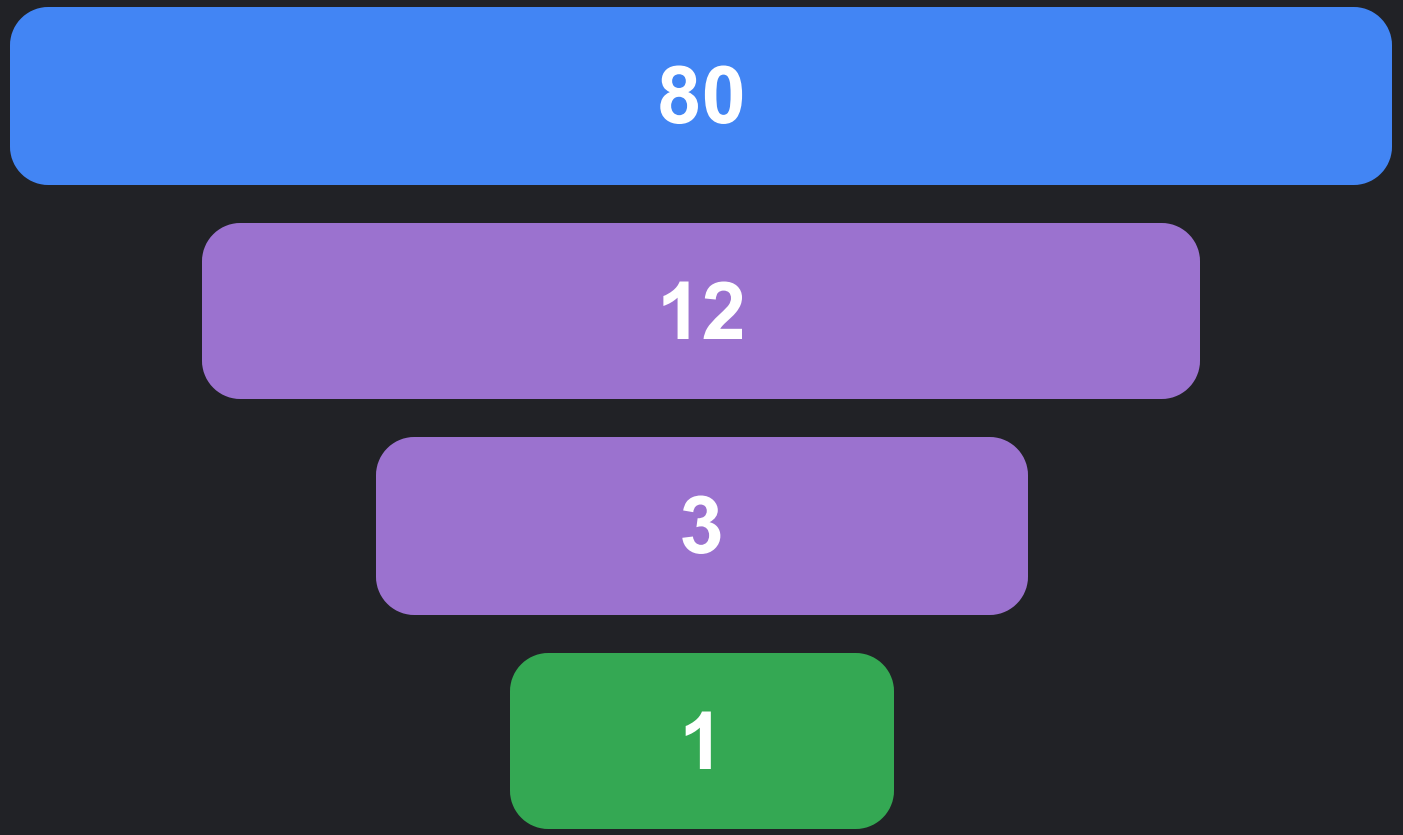
Top 40 → **3 viable**

Top 80 → **12 viable**

† Two of these three are invented teaching cases in the challenge data. From one saved run.

From 80 candidates to one call

The spinach offer: every step keeps fewer, safer options.



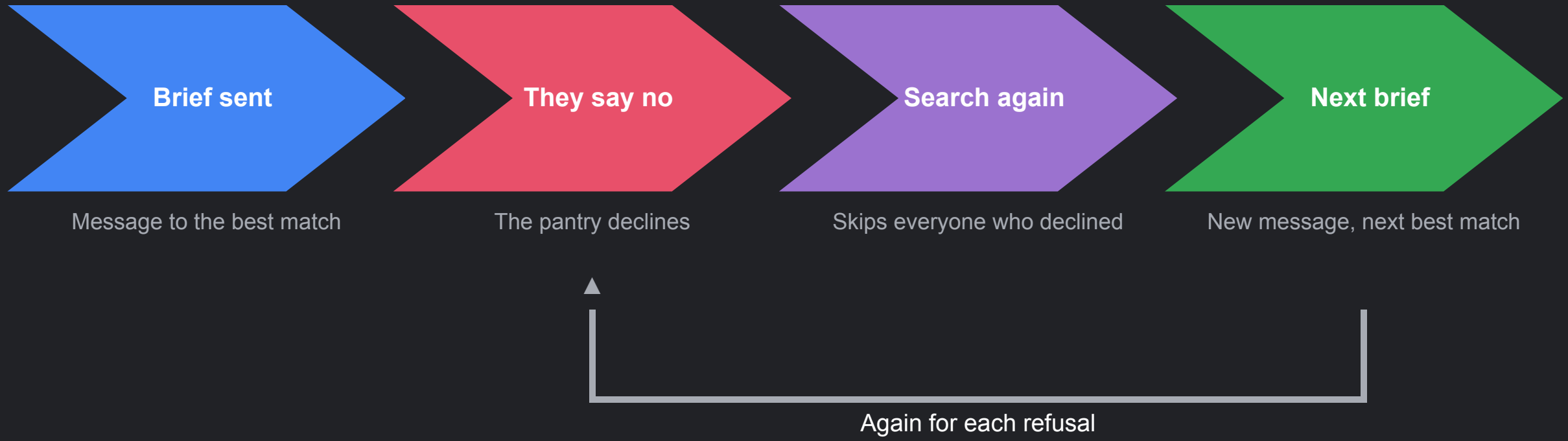
Closest by meaning
BigQuery vector search

Pass the hard filters
Cold, room and time

Ranked options
Safe margin first

Message ready to send
For the best match

When a pantry says no, the next call is ready



One pipeline for every kind of surplus

Same recipients, same search, same filters. No rulebook per track.



Grocer produce

10 crates of spinach

6 hours, must stay cold



Farm surplus

5 tons of potatoes

Needs room for pallets



Family essentials

48 cases of sealed formula

8 months to expiry



No match

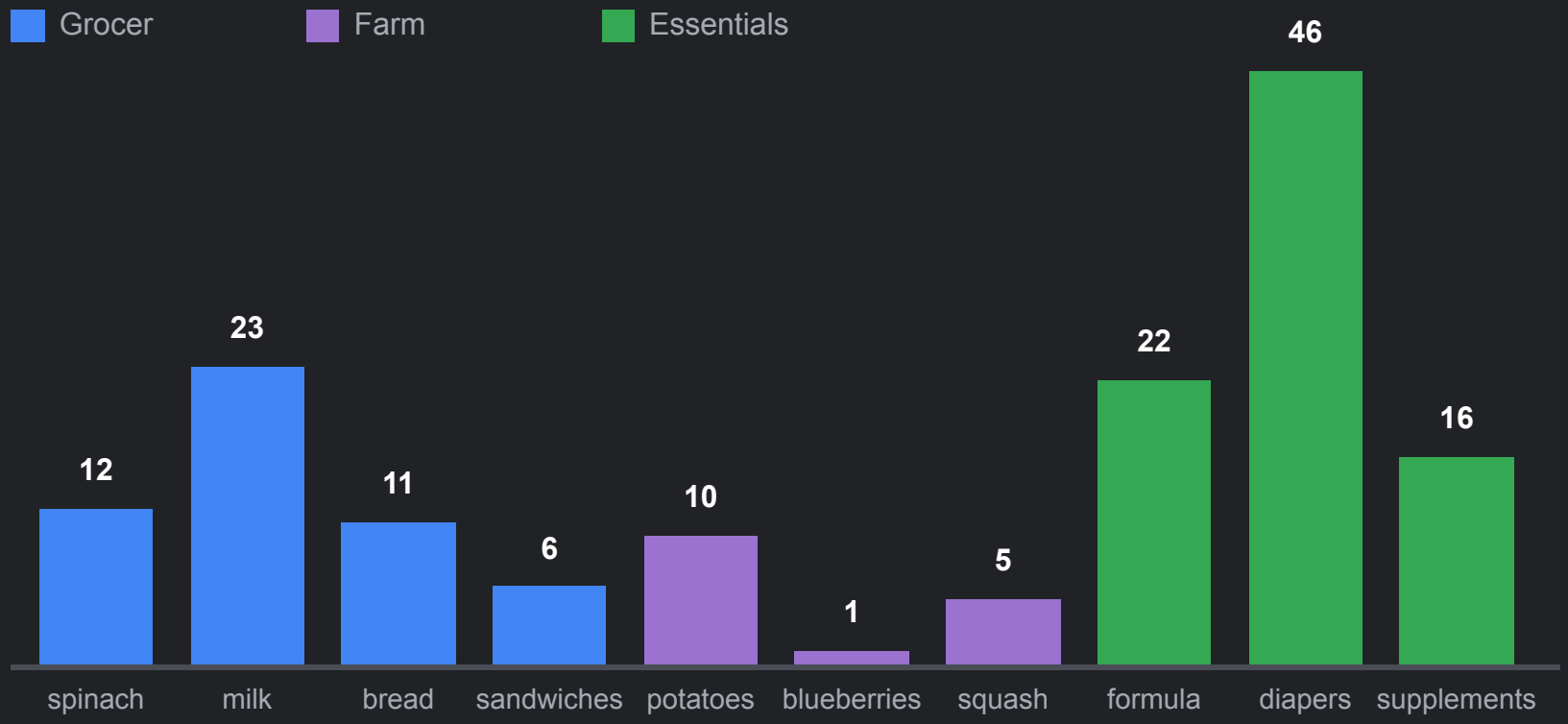
40 trays of cream pastries

Must leave in 1 hour: nobody can

When nobody can take the load, the agent says so and shows why.

Ten offers, how many pantries can take each?

Recipients that pass the hard filters, out of 80 candidates.



1 to 46

pantries can take an offer, depending on the offer

blueberries: 1. diapers: 46.

The ten sample offers in the challenge data, 183 real recipients.

What the agent decides for itself

Nobody can take it? It says so

And shows the closest rejections.

Sealed formula is not a 1-hour item

The table figure is for an opened bottle. We trust the donor’s expiry.

A kitchen fridge is not a cooler

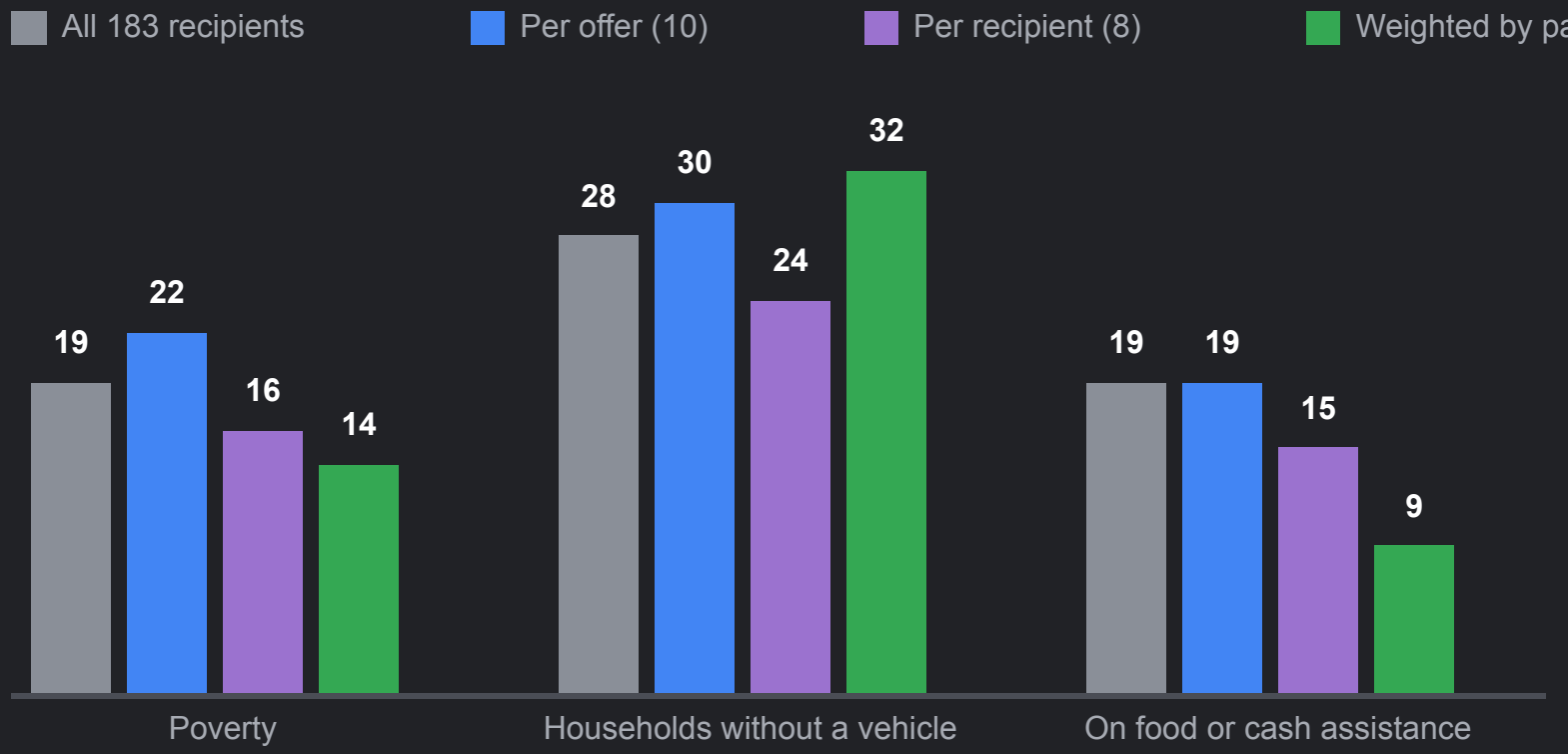
A household fridge takes at most one pallet.

Data questions use a read-only tool

It only runs SELECT. A test CREATE was refused by the server.

Per offer, close to the corpus; big loads, less assistance

Neighborhood measures where matches landed, in percent (means).



9%

assistance, weighted by estimated pallets

against 19% across all 183 recipients

Ten offers cannot separate a real skew from noise. The audit leaves out the nine invented teaching cases. Source: equity audit.

Limits today, and what is next

- **Names are real, except 9 invented test cases**
  Storage, capacity and pickup times are generated.
- **Locations are ZIP-level**
  Census figures describe the surrounding area.
- **Shelf life is consumer guidance**
  Not a commercial cold-chain guarantee.
- **One metro, ten offers**
  Chicago, ten synthetic postings.
- **The service account is broad**
  It still has the Editor role; only the data tool is read-only.
- **Filters trust the storage flags**
  Where a flag and the profile text disagree, the flag wins.

Next

- **Remember who got food**
  So one pantry is not asked every time.
- **Split a big load**
  When no single pantry can take it all.

Notebook validation: 23 of 25 checks pass; the two failures are in validation.txt.

Code and evidence: github.com/jimmycgz/challenge2-117

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Thank you

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Backup: the full diagram

